Artificial Intelligence & Machine Learning – Task 5

Ensemble Learning, Model Optimization & Real-World ML Pipeline

Objective: Build and compare classification models, optimize Random Forest using GridSearchCV, evaluate the final optimized model, perform cross-validation, and justify the final model choice.

This report follows the structure and requirements of the uploaded Task-5 assignment, which specifies Logistic Regression,
Random Forest, Gradient Boosting, GridSearchCV, model comparison, and final model selection. filecite turn0file0

1. Dataset

The recommended scikit-learn Breast Cancer dataset was used. It contains 569 samples and 30 numerical features. The target has two classes: malignant and benign. An 80:20 stratified train-test split was used, producing 455 training samples and 114 testing samples. StandardScaler was applied to Logistic Regression; the tree-based models were trained without scaling.

Property	Value
Dataset	Breast Cancer (scikit-learn)
Total samples	569
Features	30
Training samples	455
Testing samples	114
Target classes	Malignant / Benign
Test size	20%
Random state	42

2. Baseline Model – Logistic Regression

Logistic Regression was selected as the baseline classifier. Because it is sensitive to feature scale, the training data were standardized before fitting. The model was trained with max_iter=1000. On the held-out test set, it achieved 98.25% accuracy, 98.61% precision, 98.61% recall, and 98.61% F1-score.

3. Random Forest Model

Random Forest was used as the main bagging-based ensemble model. It combines multiple decision trees and introduces randomness in both samples and features. The baseline Random Forest achieved 95.61% accuracy, 95.89% precision,
97.22% recall, and 96.55% F1-score.

4. Gradient Boosting Model

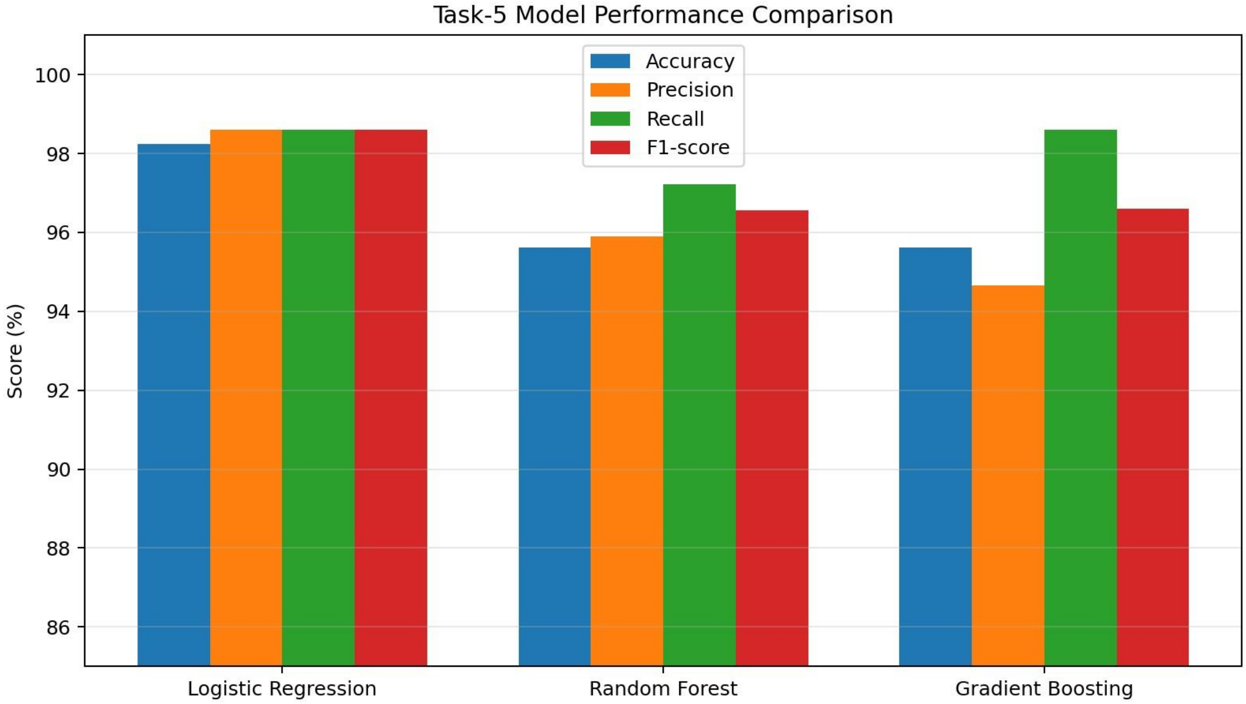
Gradient Boosting was used as the boosting model. It builds weak learners sequentially, with later learners focusing on errors made by earlier learners. It achieved 95.61% accuracy, 94.67% precision, 98.61% recall, and 96.60% F1-score.

5. Model Comparison

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	98.25%	98.61%	98.61%	98.61%
Random Forest	95.61%	95.89%	97.22%	96.55%
Gradient Boosting	95.61%	94.67%	98.61%	96.60%

Observation: Logistic Regression produced the best overall hold-out performance. Gradient Boosting had the same recall as Logistic Regression but lower precision and F1-score.

6. Visualize Model Comparison



The chart compares Accuracy, Precision, Recall and F1-score for all three classifiers. The scores show that Logistic Regression is consistently the strongest model on the test set.

7. Hyperparameter Tuning Using GridSearchCV

GridSearchCV was applied to Random Forest to systematically test combinations of n_estimators and max_depth. The search used 3-fold cross-validation and F1-score as the optimization metric. This follows the GridSearchCV requirement in the Task-5 brief. fileciteturn0file0

Parameter grid: n_estimators = [50, 100]; max_depth = [3, 5, 7]; cross-validation = 3 folds; scoring = F1.

8. Best Hyperparameters

Best parameters: n_estimators = 100, max_depth = 5
Best mean 3-fold CV F1-score: 0.9633

9. Complete GridSearchCV Output

n_estimators	max_depth	Mean CV F1	Std CV F1	Rank
100.0	5.0	0.9633	0.0074	1
50.0	7.0	0.9632	0.0131	2
100.0	7.0	0.9615	0.0152	3
100.0	3.0	0.9598	0.0111	4
50.0	5.0	0.9598	0.0111	4
50.0	3.0	0.9596	0.0113	6

The configuration with 100 trees and max_depth=5 achieved the highest mean cross-validation F1-score (0.9633) and was therefore selected as the optimized Random Forest.

10. Final Optimized Random Forest

The best estimator returned by GridSearchCV was refitted/used as the final optimized Random Forest. Its configuration is RandomForestClassifier(n_estimators=100, max_depth=5, random_state=42).

11. Final Evaluation Metrics

Metric	Optimized Random Forest
Accuracy	95.61%
Precision	95.89%
Recall	97.22%
F1-score	96.55%

12. Final Classification Report

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FINAL CLASSIFICATION REPORT
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              precision    recall  f1-score   support

   malignant       0.95      0.93      0.94        42
      benign       0.96      0.97      0.97        72

    accuracy                           0.96       114
   macro avg       0.96      0.95      0.95       114
weighted avg       0.96      0.96      0.96       114
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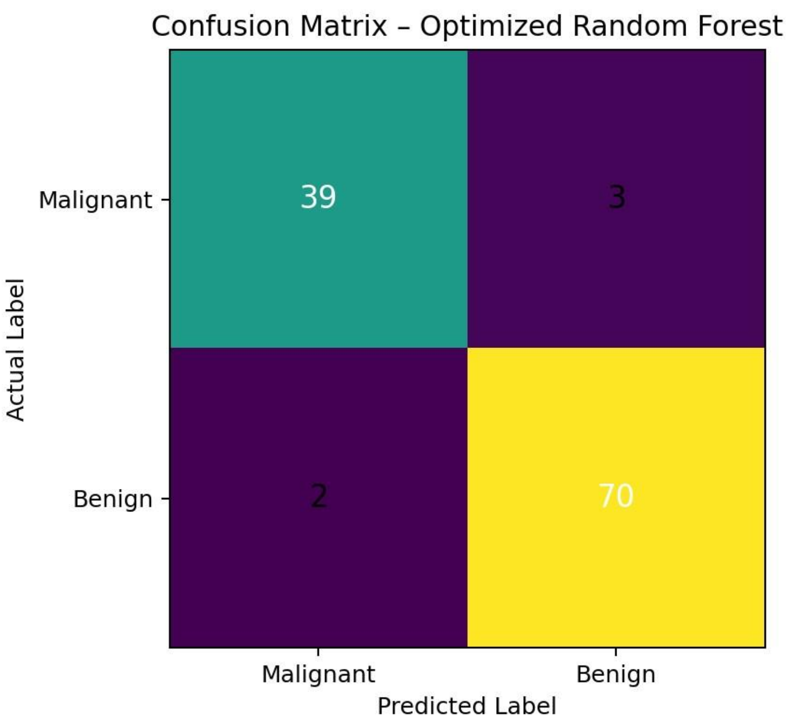
13. Confusion Matrix



The optimized Random Forest produced the confusion matrix [[39, 3], [2, 70]]. This means 39 malignant cases and 70 benign cases were classified correctly, while 3 malignant cases were predicted as benign and 2 benign cases were predicted as malignant.

14. 5-Fold Cross-Validation Stability Check

A stratified 5-fold cross-validation check was performed on the training data to assess reliability and variation across folds.

Model	Mean Accuracy	Std Accuracy	Mean F1	Std F1
Logistic Regression	0.9780	0.0098	0.9825	0.0078
Random Forest	0.9626	0.0179	0.9699	0.0146
Gradient Boosting	0.9560	0.0120	0.9649	0.0095

Logistic Regression had the highest mean cross-validation accuracy (0.9780) and mean F1-score (0.9825), while also showing relatively small fold-to-fold variation. Random Forest was stable but below Logistic Regression on both mean accuracy and mean F1-score.

15. Final Model Selection

Final selected model: Logistic Regression. The selection is based on the combined criteria of performance, stability and interpretability requested in the Task-5 assignment. Logistic Regression achieved the highest test Accuracy (98.25%), Precision (98.61%), Recall (98.61%) and F1-score (98.61%). It also achieved the strongest 5-fold cross-validation mean performance. Therefore, it is the most appropriate final model for this particular dataset.

Best optimized ensemble: Random Forest with n_estimators=100 and max_depth=5. This model satisfies the ensemble-learning and GridSearchCV optimization requirement, even though its final hold-out performance is lower than Logistic Regression.

16. Final Conclusion

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FINAL CONCLUSION
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The three classification models were successfully trained and compared.

1. Logistic Regression was used as the baseline model.
2. Random Forest was used as an ensemble learning model.
3. Gradient Boosting was used as a boosting model.
4. GridSearchCV was applied to Random Forest for hyperparameter optimization.
5. The best Random Forest parameters were identified using 3-fold cross-validation.
6. Accuracy, Precision, Recall and F1-score were used for evaluation.

FINAL MODEL DECISION:
Logistic Regression is selected as the final model for this dataset because
it achieved the strongest overall test performance and provides good
interpretability.

BEST OPTIMIZED ENSEMBLE:
Random Forest with the best hyperparameters obtained from GridSearchCV.

This completes Task 5.
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